DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

PROJECT TITLE-FLIPKART E-COMMERCE PLATFORM ANALYSIS

SKILL WEAK -8

1. Combining Multiple Tables

This query combines the Customer, Orders, Order_Details, and Product tables to display complete order information.

2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.

2.1 Customer and Orders

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT,
  7      o.ORDER_STATUS
  8  FROM Customer c
  9  INNER JOIN Orders o
 10      ON c.ID = o.CUSTOMER_ID
 11  ORDER BY o.ORDER_ID;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         1 Renu                        401 20-OCT-26        55000
Shipped

         2 Arjun                       402 21-OCT-26        25000
Placed

         3 Priya                       403 22-OCT-26         1598
Delivered


        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         4 Rahul                       404 23-OCT-26         2500
Shipped

         5 Keerthana                   405 24-OCT-26          650
Placed
```

2.2 Orders and Payment

```
SQL> SELECT
  2      o.ORDER_ID,
  3      o.ORDER_DATE,
  4      o.TOTAL_AMOUNT,
  5      o.ORDER_STATUS,
  6      p.PAYMENT_ID,
  7      p.PAYMENT_MODE,
  8      p.PAYMENT_AMOUNT,
  9      p.PAYMENT_STATUS
 10  FROM Orders o
 11  INNER JOIN Payment p
 12      ON o.ORDER_ID = p.ORDER_ID
 13  ORDER BY o.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    PAYMENT_ID
---------- --------- ------------ ------------------------------ ----------
PAYMENT_MODE                    PAYMENT_AMOUNT PAYMENT_STATUS
------------------------------ -------------- ------------------------------
       401 20-OCT-26        55000 Shipped                                601
UPI                                        55000 Completed

       402 21-OCT-26        25000 Placed                                 602
Card                                       25000 Completed

       403 22-OCT-26         1598 Delivered                              603
Cash                                        1598 Completed


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    PAYMENT_ID
---------- --------- ------------ ------------------------------ ----------
PAYMENT_MODE                    PAYMENT_AMOUNT PAYMENT_STATUS
------------------------------ -------------- ------------------------------
       404 23-OCT-26         2500 Shipped                                604
Net Banking                                2500 Completed

       405 24-OCT-26          650 Placed                                 605
UPI                                         650 Pending
```

2.3 Product and Order_Details

```
SQL> SELECT
  2      p.PRODUCT_ID,
  3      p.PRODUCT_NAME,
  4      p.PRICE,
  5      od.ORDER_ID,
  6      od.QUANTITY,
  7      od.UNIT_PRICE,
  8      od.QUANTITY * od.UNIT_PRICE AS ITEM_TOTAL
  9  FROM Product p
 10  INNER JOIN Order_Details od
 11      ON p.PRODUCT_ID = od.PRODUCT_ID
 12  ORDER BY od.ORDER_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ---------- ---------- ----------
       101
Laptop
     55000        401          1      55000      55000

       102
Smartphone
     25000        402          1      25000      25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ---------- ---------- ----------

       103
T-Shirt
       799        403          2        799       1598

       104
Mixer Grinder

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ---------- ---------- ----------
      2500        404          1       2500       2500

       105
Python Programming Book
       650        405          1        650        650
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.

3.1 Display All Customers and Their Orders

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT,
  7      o.ORDER_STATUS
  8  FROM Customer c
  9  LEFT JOIN Orders o
 10      ON c.ID = o.CUSTOMER_ID
 11  ORDER BY c.ID;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         1 Renu                        401 20-OCT-26        55000
Shipped

         2 Arjun                       402 21-OCT-26        25000
Placed

         3 Priya                       403 22-OCT-26         1598
Delivered


        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         4 Rahul                       404 23-OCT-26         2500
Shipped

         5 Keerthana                   405 24-OCT-26          650
Placed
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.

4.1 Display All Orders and Customer Details

```
SQL> SELECT
  2      o.ORDER_ID,
  3      o.ORDER_DATE,
  4      o.TOTAL_AMOUNT,
  5      o.ORDER_STATUS,
  6      c.ID AS CUSTOMER_ID,
  7      c.NAME AS CUSTOMER_NAME,
  8      c.AGE,
  9      c.ADDRESS
 10  FROM Orders o
 11  LEFT JOIN Customer c
 12      ON o.CUSTOMER_ID = c.ID
 13  ORDER BY o.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                  CUSTOMER_ID
---------- --------- ------------ ------------------------------ -----------
CUSTOMER_NAME              AGE ADDRESS
-------------------- ---------- --------------------
       401 20-OCT-26        55000 Shipped                                  1
Renu                         19 Chennai

       402 21-OCT-26        25000 Placed                                   2
Arjun                        20 Tambaram

       403 22-OCT-26         1598 Delivered                                3
Priya                        19 Vandalur


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                  CUSTOMER_ID
---------- --------- ------------ ------------------------------ -----------
CUSTOMER_NAME              AGE ADDRESS
-------------------- ---------- --------------------
       404 23-OCT-26         2500 Shipped                                  4
Rahul                        21 Chengalpattu

       405 24-OCT-26          650 Placed                                   5
Keerthana                    20 Guduvanchery
```

5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT
  2      o.ORDER_ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_DATE,
  5      p.PRODUCT_NAME,
  6      od.QUANTITY,
  7      od.UNIT_PRICE,
  8      od.QUANTITY * od.UNIT_PRICE AS ITEM_TOTAL,
  9      o.TOTAL_AMOUNT,
 10      o.ORDER_STATUS
 11  FROM Orders o
 12  JOIN Customer c
 13      ON o.CUSTOMER_ID = c.ID
 14  JOIN Order_Details od
 15      ON o.ORDER_ID = od.ORDER_ID
 16  JOIN Product p
 17      ON od.PRODUCT_ID = p.PRODUCT_ID
 18  ORDER BY o.ORDER_ID;

  ORDER_ID CUSTOMER_NAME        ORDER_DAT
---------- -------------------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------

       401 Renu                 20-OCT-26
Laptop
         1      55000      55000        55000 Shipped

       402 Arjun                21-OCT-26
Smartphone
         1      25000      25000        25000 Placed

  ORDER_ID CUSTOMER_NAME        ORDER_DAT
---------- -------------------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------

       403 Priya                22-OCT-26
T-Shirt
         2        799       1598         1598 Delivered

       404 Rahul                23-OCT-26
Mixer Grinder

  ORDER_ID CUSTOMER_NAME        ORDER_DAT
---------- -------------------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------
         1       2500       2500         2500 Shipped

       405 Keerthana            24-OCT-26
Python Programming Book
         1        650        650          650 Placed
```

6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT,
  7      o.ORDER_STATUS
  8  FROM Customer c
  9  INNER JOIN Orders o
 10      ON c.ID = o.CUSTOMER_ID
 11  ORDER BY c.ID, o.ORDER_DATE;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
--------------------------------
         1 Renu                        401 20-OCT-26        55000
Shipped

         2 Arjun                       402 21-OCT-26        25000
Placed

         3 Priya                       403 22-OCT-26         1598
Delivered


        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
--------------------------------
         4 Rahul                       404 23-OCT-26         2500
Shipped

         5 Keerthana                   405 24-OCT-26          650
Placed
```

7. Customer Purchase History with Payment

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT,
  7      o.ORDER_STATUS,
  8      p.PAYMENT_ID,
  9      p.PAYMENT_MODE,
 10      p.PAYMENT_AMOUNT,
 11      p.PAYMENT_STATUS
 12  FROM Customer c
 13  JOIN Orders o
 14      ON c.ID = o.CUSTOMER_ID
 15  LEFT JOIN Payment p
 16      ON o.ORDER_ID = p.ORDER_ID
 17  ORDER BY c.ID, o.ORDER_DATE;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS                    PAYMENT_ID PAYMENT_MODE
------------------------------ ---------- ------------------------------
PAYMENT_AMOUNT PAYMENT_STATUS
-------------- ------------------------------
         1 Renu                        401 20-OCT-26        55000
Shipped                               601 UPI
         55000 Completed

         2 Arjun                       402 21-OCT-26        25000
Placed                                602 Card
         25000 Completed

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS                    PAYMENT_ID PAYMENT_MODE
------------------------------ ---------- ------------------------------
PAYMENT_AMOUNT PAYMENT_STATUS
-------------- ------------------------------

         3 Priya                       403 22-OCT-26         1598
Delivered                             603 Cash
          1598 Completed

         4 Rahul                       404 23-OCT-26         2500
Shipped                               604 Net Banking

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS                    PAYMENT_ID PAYMENT_MODE
------------------------------ ---------- ------------------------------
PAYMENT_AMOUNT PAYMENT_STATUS
-------------- ------------------------------
          2500 Completed

         5 Keerthana                   405 24-OCT-26          650
Placed                                605 UPI
           650 Pending
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      p.PRODUCT_NAME,
  7      od.QUANTITY,
  8      od.UNIT_PRICE,
  9      od.QUANTITY * od.UNIT_PRICE AS ITEM_TOTAL,
 10      o.TOTAL_AMOUNT,
 11      o.ORDER_STATUS
 12  FROM Customer c
 13  JOIN Orders o
 14      ON c.ID = o.CUSTOMER_ID
 15  JOIN Order_Details od
 16      ON o.ORDER_ID = od.ORDER_ID
 17  JOIN Product p
 18      ON od.PRODUCT_ID = p.PRODUCT_ID
 19  ORDER BY o.ORDER_ID;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT
---------- -------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------
         1 Renu                        401 20-OCT-26
Laptop
         1      55000      55000        55000 Shipped

         2 Arjun                       402 21-OCT-26
Smartphone
         1      25000      25000        25000 Placed

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT
---------- -------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------

         3 Priya                       403 22-OCT-26
T-Shirt
         2        799       1598         1598 Delivered

         4 Rahul                       404 23-OCT-26
Mixer Grinder

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT
---------- -------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------
         1       2500       2500         2500 Shipped

         5 Keerthana                   405 24-OCT-26
Python Programming Book
         1        650        650          650 Placed
```

9. Payment Transaction Report

```
SQL> SELECT
  2      o.ORDER_ID,
  3      o.ORDER_DATE,
  4      o.TOTAL_AMOUNT,
  5      o.ORDER_STATUS,
  6      p.PAYMENT_ID,
  7      p.PAYMENT_MODE,
  8      p.PAYMENT_DATE,
  9      p.PAYMENT_AMOUNT,
 10      p.PAYMENT_STATUS
 11  FROM Orders o
 12  INNER JOIN Payment p
 13      ON o.ORDER_ID = p.ORDER_ID
 14  ORDER BY o.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    PAYMENT_ID
---------- --------- ------------ ------------------------------ ----------
PAYMENT_MODE                   PAYMENT_D PAYMENT_AMOUNT
------------------------------ --------- --------------
PAYMENT_STATUS
------------------------------
       401 20-OCT-26        55000 Shipped                             601
UPI                            20-OCT-26          55000
Completed

       402 21-OCT-26        25000 Placed                              602
Card                           21-OCT-26          25000
Completed

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    PAYMENT_ID
---------- --------- ------------ ------------------------------ ----------
PAYMENT_MODE                   PAYMENT_D PAYMENT_AMOUNT
------------------------------ --------- --------------
PAYMENT_STATUS
------------------------------

       403 22-OCT-26         1598 Delivered                           603
Cash                           22-OCT-26           1598
Completed

       404 23-OCT-26         2500 Shipped                             604
Net Banking                    23-OCT-26           2500

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    PAYMENT_ID
---------- --------- ------------ ------------------------------ ----------
PAYMENT_MODE                   PAYMENT_D PAYMENT_AMOUNT
------------------------------ --------- --------------
PAYMENT_STATUS
------------------------------
Completed

       405 24-OCT-26          650 Placed                              605
UPI                            24-OCT-26            650
Pending
```